Project Plan & Pattern Mapping Document

Team: Sangre y Oro

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1. Project Scope

This project will deliver a web-based matching platform connecting Erasmus and local students based on shared interests, hobbies, and language goals.

- **Included**: User registration, interest/skill-based profiling, configurable matching systems, event creation and participation, user notifications, and a multi-level inappropriate content reporting system. The user interface will be developed in Turkish, English, and Spanish. eduroam authentication and translation APIs will be simulated using mock implementations via Adapters.
- **Not Included**: The development of native iOS or Android mobile applications, deployment into a real production environment with live eduroam or commercial translation API billing, real-time chat functionality, payment gateways, and formal legal GDPR/KVKK auditing.

2. Project Organization

- **Alberto Alvarez Polo:** Responsible for the Project Definition Document and the Software Architecture Overview.
- **Alejandro Blas Piris:** Responsible for the Project Plan Document and the Pattern-Use Case Mapping Matrix.
- **Carlos Molina Perez:** Responsible for the Software Requirements Specification (SRS) and the final integration of the architecture and pattern matrix into the SRS.

3. Objectives

The management and execution objectives for this project are to:

- Establish a structured, predictable timeline ensuring all Project Assignments (PA1, PA2, PA3) and the final source code are delivered on or before the deadlines.
- Ensure clear tracking of individual contributions via GitHub for grading transparency.
- Maintain an organized layered architecture to allow concurrent development by all three team members without codebase conflicts.

4. Key Phases and Timeline

- **Phase 1 - Requirements & Initial Design (PA1):** Gather requirements, define architecture, select patterns. Delivery of PA1 (Project Definition, Plan, SRS).
- **Phase 2 - Detailed Design (PA2):** In-depth application of patterns to one selected use case. Presentation in the Design Pattern Workshop (PA2).
- **Phase 3 - Final Architecture & Technologies (PA3):** Finalization of project design, technologies, and integration strategies (PA3).
- **Phase 4 - Implementation & Testing:** Coding phase. Implementing UC1-UC4, Layered Architecture, and the five target Design Patterns.

- **Phase 5 - Closure:** Final source code push to GitHub. Live presentation and working demonstration of the platform.

5. Resource Planning

- **Development Environment**: Web stack (HTML/CSS/JS) for the Presentation Layer, and an Object-Oriented backend language to effectively implement the Business Layer patterns. A relational database will be used for the Data Access Layer.
- **Collaboration Tools**: Git/GitHub for version control and issue tracking. Google Drive for document aggregation and final PDF submissions to the instructor.

6. Risk Management

- **Schedule Delays:** Mitigated by setting internal deadlines prior to the official submission date.
- **Technical Difficulties:** By isolating development into independent services (UserService, EventService, etc.), issues in one area will not block the progress of others.
- **Team Absence (Illness):** Code and documents will be centralized so any member can continue the work if needed.

7. Communication Plan

The team will maintain continuous daily communication through a dedicated WhatsApp group for immediate query resolution. Additionally, a formal weekly synchronization meeting will be held to review GitHub progress, align on architecture decisions, and plan upcoming tasks.

8. Change Management Plan

Any proposed changes to the project scope, core use cases, or assigned design patterns must be raised during the weekly synchronization meeting. A change will only be adopted if all three team members agree it is feasible. Accepted changes must be immediately reflected in updates to the SRS.

9. Budget Plan

Project effort will be distributed to balance planning and execution:

- 30% - Design & Documentation: PA1, PA2, and PA3 deliverables, ensuring strong architectural foundations.
- 50% - Implementation: Writing code for the core Use Cases (UC1-UC4) and applying patterns.
- 20% - Testing & Closure: End-to-end testing, bug fixing, and preparing the final demonstration.

10. Acceptance Tests and Acceptance Criteria

The project will be considered successful if it meets the following criteria prior to the final demonstration:

- **Functional End-to-End**: All four core use cases (UC1-UC4) execute without critical errors in a live demo.
- **Traceable Design**: Every implemented use case explicitly utilizes at least one documented design pattern, clearly identifiable in the source code.
- **Usability Benchmark**: A user can complete registration and receive their first match in under 5 minutes.
- **Multilingual Support**: The interface switches successfully between Turkish, English, and Spanish.

11. Task Matrix

Task	Responsible
Project Scope, Organization & Objectives	Alejandro Blas Piris
Timeline, Resources, Risk & Communication	Alejandro Blas Piris
Change Management, Budget & Acceptance Tests	Alejandro Blas Piris
Pattern-Use Case Mapping Matrix	Alejandro Blas Piris